January 23, 2026

Let's recap on the last lecture by asking when two lines intersect

e.g. 1.2.6 $\begin{cases} x = t+1 \\ y = 5t+6 \\ z = -2t \end{cases}$ and $\begin{cases} x = 3t-3 \\ y = t \\ z = t+1 \end{cases}$

line 1: $\vec{r}_1(t) = (t+1, 5t+6, -2t)$; line 2: $\vec{r}_2(t) = (3t-3, t, t+1)$

the lines intersect if $\vec{r}_1(t_1) = \vec{r}_2(t_2)$ (for some $t_1, t_2 \in \mathbb{R}$)

① $t_1 + 1 = 3t_2 - 3$

② $5t_1 + 6 = t_2$

③ $-2t_1 = t_2 + 1$

$-2t_1 = t_2 + 1 = (5t_1 + 6) + 1$

$-2t_1 = 5t + 7$

$-7 = 7t_1$, $t_1 = -1$ $t_2 = 5(-1) + 6$

$t_1 = -1$, $t_2 = 1$

we need to go back and check if t_1 and t_2 satisfy all the equation

① $-1+1 = 0$ $3(1) - 3 = 0$ ✓

② $5(-1) + 6 = 1$ $t_2 = 1$ ✓

③ $-2(-1) = 2$ $1+1 = 2$ ✓

If trying to solve the system of equation we get a contradiction (e.g. $0 = 4$) then the system is inconsistent and the two lines never meet

If we a unique solution for t_1 and t_2 the lines intersect at just one point

If we get a tautology (e.g. $t = t$, $0 = 0$), then the two lines are equal.

1.3 The Dot product (also called inner product or scalar product)

def: Let $\vec{a} = (a_1, a_2)$ and $\vec{b} = (b_1, b_2)$, we define the dot product of $\vec{a}$ and $\vec{b}$ as $\vec{a}\cdot\vec{b} = a_1 \cdot b_1 + a_2 \cdot b_2$.

in $\mathbb{R}^3$: $\vec{a} = (a_1, a_2, a_3)$, $\vec{b} = (b_1, b_2, b_3)$ and $\vec{a}\cdot\vec{b} = a_1b_1 + a_2b_2 + a_3b_3$

(we can do dot product between two vectors in $\mathbb{R}^n$: $\vec{a}\cdot\vec{b} = a_1 b_1 + \dots + a_n b_n$)

Examples:

- $\underbrace{(3,1,2)}_{3\hat{\imath} + \hat{\jmath} + 2\hat{k}} \cdot (1,-1,3) = (3)(1) + (1)(-1) + (2)(3)$
  $= 3 - 1 + 6 = 8$

- $(2\hat{\imath} + \hat{\jmath} - 2\hat{k}) \cdot (3\hat{\imath} - \hat{\jmath}) = (2)(3) + (1)(-1) + (-2)(0)$
  $= 6 - 1 + 0 = 5$

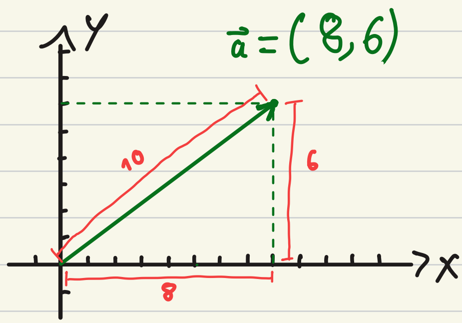
observe: $\hat{\imath}\cdot\hat{\imath} = 1$, $\hat{\jmath}\cdot\hat{\jmath} = 1$, $\hat{k}\cdot\hat{k} = 1$, $\hat{\imath}\cdot\hat{\jmath} = 0$, $\hat{\jmath}\cdot\hat{k} = 0$, $\hat{k}\cdot\hat{\imath} = 0$

properties of dot products: Let $\vec{a}, \vec{b}, \vec{c} \in \mathbb{R}^2$ (or $\mathbb{R}^3$, or $\mathbb{R}^n$) and $k \in \mathbb{R}$:

1. $\vec{a}\cdot\vec{a} \geq 0$ (always!), and $\vec{a}\cdot\vec{a} = 0$ if and only if $\vec{a} = \vec{0}$
2. $\vec{a}\cdot\vec{b} = \vec{b}\cdot\vec{a}$ (commutativity)
3. $\vec{a}\cdot(\vec{b}+\vec{c}) = \vec{a}\cdot\vec{b} + \vec{a}\cdot\vec{c}$ (distributivity)
4. $k(\vec{a}\cdot\vec{b}) = (k\vec{a})\cdot\vec{b} = \vec{a}\cdot(k\vec{b})$

def: If $\vec{a} = (a_1, a_2, a_3)$, we define the length (or norm, or magnitude) of $\vec{a}$ as

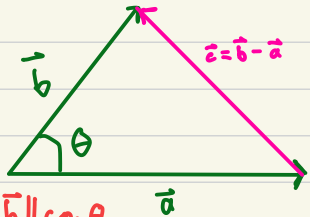
$$\|\vec{a}\| = \sqrt{a_1^2 + a_2^2 + a_3^2}$$



observe: $\|\vec{a}\| = \sqrt{\vec{a}\cdot\vec{a}}$

Geometric interpretation of inner product

Theorem: $\vec{a}\cdot\vec{b} = \|\vec{a}\|\,\|\vec{b}\|\cos\theta.$



idea: Law of cosines

$$\|\vec{c}\|^2 = \|\vec{a}\|^2 + \|\vec{b}\|^2 - 2\|\vec{a}\|\,\|\vec{b}\|\cos\theta$$

$$\vec{c}\cdot\vec{c} = (\vec{b}-\vec{a})\cdot(\vec{b}-\vec{a}) = \vec{b}\cdot(\vec{b}-\vec{a}) - \vec{a}\cdot(\vec{b}-\vec{a})$$

$$= \vec{b}\cdot\vec{b} - \vec{b}\cdot\vec{a} - \vec{a}\cdot\vec{b} + \vec{a}\cdot\vec{a} = \cancel{\vec{a}\cdot\vec{a}} + \cancel{\vec{b}\cdot\vec{b}} - 2\vec{a}\cdot\vec{b} \quad \text{LHS}$$

$$\| \quad \cancel{\vec{a}\cdot\vec{a}} + \cancel{\vec{b}\cdot\vec{b}} - 2\sqrt{\vec{a}\cdot\vec{a}}\sqrt{\vec{b}\cdot\vec{b}}\cos\theta \quad \text{RHS}$$

$$\vec{a}\cdot\vec{b} = \sqrt{\vec{a}\cdot\vec{a}}\sqrt{\vec{b}\cdot\vec{b}}\cos\theta$$

Theorem (angle between two vectors): $\theta = \cos^{-1}\left(\frac{\vec{a}\cdot\vec{b}}{\|\vec{a}\|\,\|\vec{b}\|}\right)$ $(\theta\in[0,\pi])$.

1.3 Example 2 $\quad \vec{a} = \hat{\imath}+\hat{\jmath} \quad \vec{b} = \hat{\jmath}-\hat{k} \quad$ angle between $\vec{a}$ and $\vec{b}$?

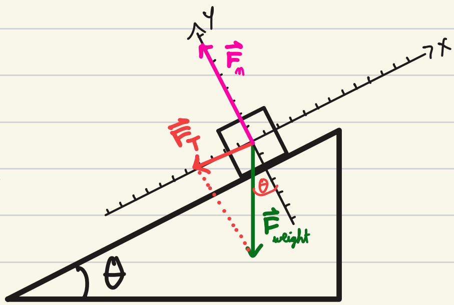
$$\vec{a}\cdot\vec{b} = (\hat{\imath}+\hat{\jmath})\cdot(\hat{\jmath}-\hat{k}) = \hat{\imath}\cdot(\hat{\jmath}-\hat{k}) + \hat{\jmath}\cdot(\hat{\jmath}-\hat{k}) = \hat{\imath}\cdot\hat{\jmath} - \hat{\imath}\cdot\hat{k} + \hat{\jmath}\cdot\hat{\jmath} - \hat{\jmath}\cdot\hat{k}$$
$$= 0 - 0 + 1 = 1$$

$$\|\vec{a}\| = \sqrt{(1,1,0)\cdot(1,1,0)} = \sqrt{(1)(1)+(1)(1)} = \sqrt{2} \qquad \|\vec{b}\| = \sqrt{(0,1,-1)\cdot(0,1,-1)} = \sqrt{2}$$

$$\cos\theta = \frac{\vec{a}\cdot\vec{b}}{\|\vec{a}\|\,\|\vec{b}\|} = \frac{1}{\sqrt{2}\sqrt{2}} = \frac{1}{2} \quad \rightsquigarrow \quad \theta = \pi/3$$

What does it mean if $\vec{a}\cdot\vec{b}=0$? $\quad \vec{a}$ and $\vec{b}$ are perpendicular

Vector projection

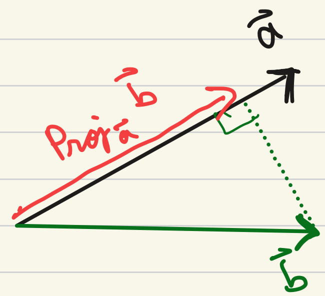


e.g.: $\vec{F}_T = \vec{F}_w + \vec{F}_n = m\vec{a}$

$$\|\vec{F}_T\| = \|\vec{F}_w\|\ \sin\theta$$

The projection of $\vec{b}$ in the direction of $\vec{a}$ is

$$\text{proj}_{\vec{a}}\, \vec{b} = \left(\frac{\vec{a}\cdot\vec{b}}{\vec{a}\cdot\vec{a}}\right)\vec{a}$$



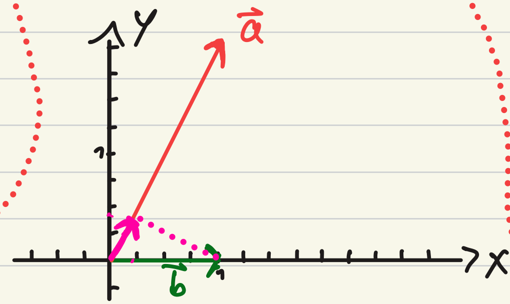
1.3. Example 4

$\vec{b} = (1,0) \qquad \vec{a} = (1,2)$

Projection of $\vec{b}$ in the direction of $\vec{a}$?

$$\text{proj}_{\vec{b}}\,\vec{a} = \frac{\vec{a}\cdot\vec{b}}{\vec{a}\cdot\vec{a}}\,\vec{a}$$

mistake : the order is swapped



$\vec{a}\cdot\vec{b} = (1)(1) + (2)(0) = 1$

$\vec{a}\cdot\vec{a} = (1)(1) + (2)(2) = 5$

$$\text{proj}_{\vec{a}}\,\vec{b} = \frac{\vec{a}\cdot\vec{b}}{\vec{a}\cdot\vec{a}}\,\vec{a} = \frac{1}{5}(1,2) = \left(\frac{1}{5}, \frac{2}{5}\right)$$

We are looking for how much of $\vec{b}$ points in the direction of $\vec{a}$, that means we apply the projection in the direction of $\vec{a}$ to the vector $\vec{b}$

$\text{proj}_{\vec{a}}\,\vec{b}$